The card

The design of a screener card: what is on it, in what order, how each part is drawn, and the rules behind the drawing. Written from `buildCard()` and its helpers in `public/index.html`.

For	Read
where each number comes from	DATA-PIPELINE.md
every field, its source, and when a blank is legitimate	CARD-FIELDS.md
which of these fields are frozen into the registers	REGISTERS.md

Layout, top to bottom

```
┌──────────────────────────────────────────────────────┐
│ NVDA  TV↗  $128.40  +4.12%  ● Live          72  ☆    │  header
│ Electronic Technology · Semiconductors      B4 main  │
├──────────────────────────────────────────────────────┤
│ first seen 09:12 · 18m before open  ★ also T7        │  badges
│ [Big Move 09:12] [Pre-Mkt 08:30]  🔥 HOT  ▲▲ ALL BULL │
│ ⚡ Earnings Beat +1                                  │
├──────────────────────────────────────────────────────┤
│ ▲9MA ▲13MA ▲20MA ▲50MA ▲5MA ▲VWAP ▲PDC ▲OPEN ▲PMH ▲PML │  signal tags
│ 9>13>20>50   Q4 range   PM/ADR 1-2 · 1.34×           │
├──────────────────────────────────────────────────────┤
│ ▸ Chart                    tap to open               │  folds, each
│ ▸ Volume & Float           RVOL 6.2× · cap $3.1T · … │  with a one-line
│ ▸ Range                    W 92% · M 88% · Q 74% · Y 81% summary when
│ ▸ News · ⚡ Earnings Beat  3 stories · Nvidia beats … │  closed
│ ▸ CANSLIM · O'Neil         C A N S L I M one-liner   │
│ ▸ Themes                                             │
│ ▸ Market Context           STRONG_UP · long BULLISH · … │
│ ▸ Price                    $128.40 · gap +2.1% · vwap … │
│ ▸ EMAs                     9 … · 20 … · 50 …         │
│ ▸ ATR & Move               ADR 3.2% · move 1.40× ATR │
│ ▸ Pre-Market Range         PM/ADR 1.34× · range $3.90 │
├──────────────────────────────────────────────────────┤
│ score breakdown (opens when the score is tapped)     │
└──────────────────────────────────────────────────────┘
```

One level of folding. Every section is a fold, closed by default except where noted, and each closed fold shows its key numbers on its heading line. There is no “Details” button wrapping folds inside folds.

Header

Element	Content	Rule
Ticker	the symbol	tap opens the full chart overlay
`TV ↗`	link to the symbol on TradingView	uses `tvSymbol` (e.g. `NASDAQ:AAPL`)
Price	`$` price	plain text: a price is a fact, not a claim
Change	`+x.xx%` today	green up, red down
`Live` / `Stale`	Live = matched by a screener in the last scan; Stale = matched earlier today, still re-quoted every 5 minutes	a stale card is dimmed
Sector · Industry	TradingView labels	—
Score	0–100, or `–` when there is none	≥70 high, ≥40 mid, below low; tap opens the score breakdown. `–` on every card today (scorer off, D-D1)
Table label	e.g. `B4 main` or `B4 sub:STRONG_UP`	which scoring table produced the score; tap opens the table inspector
☆ / ★	this tool’s shortlist star	your own mark, yellow

Badges row

In this order:

1. **First seen** `HH:MM` and `Nm before open` / `at the open` / `+Nm`: the earliest moment any of this tool’s screeners matched the stock. Pre-market, early (≤ 30 min after the open) and late are styled differently.
2. **★ also T2 T7**: other tools that shortlisted the same ticker today (violet: someone else’s opinion).
3. **Screener badges**: one per screener that ever matched it today, with the time it first matched, coloured by screener. A screener that matched earlier but not in the last scan is drawn faded (`badge-past`); the tooltip gives its run window.
4. 🔥 **HOT**: its sector is hot (MARKET-TAB.md).

5. ▲▲ **ALL BULL / ▼▼ ALL BEAR**: the price is above every one of EMA 9, 13, 20, 50, SMA 5, VWAP, previous close, open and the PM low, the EMA stack is 9>13>20>50, **and** the catalyst is bullish (the mirror for bearish, with the PM high). The PM high/low on the other side and the month quarter are not part of it.
6. **Catalyst**:
   - ⚡ marks tier 1 (major), ● tier 2, · tier 3, and 📈 a technical catalyst (the tape, not a story);
   - `+N` counts the further catalysts found;
   - a stale story (over 4 days old) is dimmed;
   - the tooltip gives the tier, the detail, how many headlines corroborate it, its age, the confidence and the evidence headline.

Signal tags

One row of tags drawn from `row.signals`:

- **Ten level tags**: `9MA 13MA 20MA 50MA 5MA VWAP PDC OPEN PMH PML`.
  - ▲ green when the price is above the level, ▼ red when below.
  - The tooltip gives the distance in %.
  - A missing level means no tag.
- **Stack**: `9>13>20>50` (bull), `9<13<20<50` (bear) or `MA mixed`, with `n/3` in order.
- **Month quarter**: `Q1 range` … `Q4 range` (Q1 = bottom of the 1-month range).
- **PM/ADR band**: `<0.5`, `0.5-1`, `1-2`, `2-3`, `3+`, with the ratio.

The folds

#	Fold	Closed line	Open
1	**Chart**	tap to open	a TradingView widget (`tv.js`), daily candles, dark, no toolbar. Mounted only when the card is on screen **and** the fold is open; `⤢ full` opens the full chart. If `tv.js` did not load, it says so (D-S4)
2	**Volume & Float**	RVOL · cap · float · days to cover	RVOL, market cap, float, short %, days to cover — each coloured by magnitude (below)
3	**Range**	`W 92% · M 88% · Q 74% · Y 81%`	four bars (week = 5 sessions, month, quarter, year), each with its low, high and the price's position; all-time high as a distance (no low exists)
4	**News**	`N stories · top headline` , or `nothing in three weeks`	up to 5 newest headlines across all sources with source tag (TV, BZ = Alpaca/Benzinga, FH, GN, YH, SEC), age and link; a warning when a source is not answering; how many older stories were hidden; when it was fetched. "Load News" when not fetched yet
5	**CANSLIM · O'Neil**	the C A N S L I M one-liner	the seven letters (C, A, N, S, L, I, M), a "Not shown:" line naming each missing letter and why, relative strength (RS line, divergence), and a nested **Full tables** fold: eight quarters, the annual run, the whole base. Field by field in CARD-FIELDS.md
6	**Themes**	tap to open	theme tags (AI, Space, Nuclear …) from a ticker list and an industry list
7	**Market Context**	regime · long · short	regime with its icon, stance and colour; long, mid and short term; the sector with its bias and score; the **bias button** (below)
8	**Price**	price · gap · VWAP	price, previous close, VWAP, 5-day MA, open, gap %
9	**EMAs**	9 · 20 · 50	EMA 9, 13, 20, 50, each green above / red below the price
10	**ATR & Move**	ADR % · move × ATR	ADR $, ADR %, today's range ÷ ATR, day high/low
11	**Pre-Market Range**	PM/ADR · range	PM high, low, range, PM/ADR

The CANSLIM data is fetched by the browser when cards are shown, through the tool's `/api/market/oneil/*` routes, which ask qp.

Range rules

- **Reverse splits.** TradingView's longer-window highs are not split-adjusted. When a window's high is more than **10×** the next shorter window's high, that window and every wider one are hidden. The card says which, and why. The all-time-high distance is suppressed the same way.
- **Identical windows.** When a shorter window has exactly the same low and high as the next wider one, it is labelled `= month` (the whole month happened this week), so it does not look like a broken column.

The bias button

`⊙ AUTO (long ⚡)` → `▲ LONG` → `▼ SHORT` → back to AUTO, saved per card.

- **AUTO** shows what the desk would choose and why (DATA-PIPELINE.md, auto bias). ⚡ means a catalyst decided it.
- **(no read)** means nothing points either way.
- A manual choice is carried into scoring.

Colour language

One meaning per colour, everywhere on the card:

Role	Used for
Direction (green / red)	price relative to something: above/below a level, up/down on the day. A good score is never green.
Magnitude (one hue, 4 intensities)	how big, never whether it is good: RVOL, float, cap, short %, days to cover, PM/ADR, sector heat
Model (orange)	the score, and only the score
Someone else (violet)	another tool's shortlist
Mine (yellow)	your shortlist star
No claim (grey)	missing data, below a threshold, switched off
Furniture (blue)	links, borders, bar fills; never used to say something

Magnitude steps (present / notable / extreme):

Field	1	2	3
RVOL	≥ 1.5×	≥ 3×	≥ 5×
Float (smaller is louder)	≤ 50M	≤ 20M	≤ 10M
Market cap (smaller is louder)	≤ $2B	≤ $300M	≤ $50M
Short %	≥ 5 %	≥ 10 %	≥ 20 %
Days to cover	≥ 2	≥ 5	≥ 10

Above the cards

Control	Options
Search	ticker or sector text
Screener	all, or one screener (cards it ever matched today)
News	any · has news · no news · not checked yet
Catalyst	any · has catalyst · ⚡ major only · no catalyst
Aligned	any · ▲ fully bullish · ▼ fully bearish · either direction
Sort	score ↓ · change ↓ · RVOL ↓ · live first · newest found ↓

The score breakdown

Tap the score: the base used (e.g. B4 = 09:40 long), the entry time and bias, the market regime of this card, which table scored it (a regime-specific `sub` table, or the `main` table and why), the bias the table read, its sample count, its confidence, a bar per factor’s bucket score, and a button to open that table.